

DATA ANALYTICS CURRICULUM

Leveraging Data Analytics for Social Good



SABUDH



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About Sabudh

Sabudh Foundation was founded by Dr. Sarabjot Singh Anand in 2018, along with Dr. Sukhjit Sehra. It empowers students to apply Data Analytics to solve real-world problems and collaborate with citizens, governments, and NGOs to create meaningful societal impact.

Recognizing that successful Data Analysts must not only fulfill business needs but also continuously expand their skill set in this rapidly evolving domain, the program fosters both practical expertise and a mindset of lifelong learning in analytics.

Data Analytics

[6 Months]

Program Learning Objectives

The Data Analytics program is designed to equip students with the technical and analytical skills required to process, analyze, and interpret data for informed decision-making. The key learning objectives include:

- Gain proficiency in Python and SQL for data collection, cleaning, and transformation. Learn how to work with large datasets efficiently.
- Develop the ability to create meaningful visualizations using Power BI and Python libraries. Learn to build interactive dashboards for data-driven storytelling.
- Understand distributed computing concepts and apply them using PySpark to handle large-scale datasets.
- Apply supervised and unsupervised learning techniques to extract insights from data and make accurate predictions.
- Learn how to automate repetitive data processing tasks using Python and Dataiku to enhance productivity and efficiency.
- Work on capstone projects that simulate industry scenarios, ensuring readiness for data-driven careers in various sectors.

01 Python Programming

[8 weeks]

Course Learning Objectives

The objective of this course is to make students to

- Master the basic programming constructs in Python. Understand how string manipulation and class structures support sequence methods in Python.
- Explore advanced Python topics such as inheritance, exception handling, and working with key data structures like lists, dictionaries, and sparse matrices.
- Develop skills to handle, manipulate, and visualize various forms of data, including text, numerical data, and media types, using libraries like Pandas, Matplotlib, and Seaborn.
- Learn how to scrape data from the web, parse HTML/JSON, work with REST APIs, implement GUIs using Python, and develop robust software by applying best practices in testing and debugging.

UNIT 1: Python Basics, Modules and Packages

Python Installation, Development environment, Programming using Python, Basic input and output, Errors, Variables and assignments, Identifiers, Objects, Module basics, Math module, modules, importing specific names from a module, executing modules as scripts, Reloading modules, Packages and Standard library, Reading files, Writing files, Interacting with file systems, Binary data, Command-line arguments and files, the 'with' statement, Comma separated values files.

UNIT 2: Data Types, Operators

Numeric types including integers and floating-point numbers, representing text using strings, string basics such as indexing, slicing, and concatenation, arithmetic expressions and operator precedence, Python expressions and their evaluation, equality and relational operators, Boolean operators and expressions with short-circuit evaluation, order of evaluation of mixed operators, membership operators (in, not in), identity operators (is, is not), and practical exercises combining these concepts.

UNIT 3: Control Flow, File Handling, Error Handling

Set as basics, Common data types summary, Type conversions, Binary numbers, String formatting, If-else statement, Code blocks and indentation, Conditional expressions Loops, while loops, Counting, For loops, Counting using the range () function, while vs. for loops, Nested loops, Break and continue, String slicing, Advanced string formatting, String methods, Splitting and joining strings, the string format method, itertools, eval, exec, enumerate, zip, copy, Try – Catch Block for error handling.

UNIT 4: Data Structures: List, Tuple, Set, Dictionary, Dataframe, Array and Data Visualization

Lists, Dictionaries, Mutability and References, Tuples, Sets and Collections, Numpy array, Pandas Dataframe-Using Pandas for data manipulation, Matplotlib and Seaborn for data visualization, Plotly for interactive Visualisation, Handling common data types: Images, Pdfs, Spreadsheets and Text.

UNIT 5: Functions and Recursion, Generators and Decorators

User-defined function basics, returning values from functions, Reasons for defining functions, Function stubs, Functions: Common errors, Function arguments, First class functions, anonymous functions, map, filter and reduce, Recursion: Definition and Principle, Types of recursion, Base Case, Recursive Step, Applications, Call Stack: Understanding how the computer's call stack manages recursive function calls.

Unit 6: OOP Concepts , Classes and Inheritance

Classes: Class methods, Class and instance object types, Class constructors, Class interfaces, Class customization, Classes as user defined types, Inheritance: Base class and Derived classes concepts, use of super(), overriding and overloading concepts, accessing base class attributes, is-a versus has-a relationships, Mixing classes and multiple inheritance.

Unit 7: Memory Management in Python

Memory allocation and garbage collection, Decorators, Generators, Handling exceptions using try and except, Multiple exception handlers, raising exceptions, Exceptions with functions, using finally to clean up, Custom exception types.

Unit 8: Scientific Computing for Machine Learning

Numerical Computing using NumPy for array manipulation, scientific computing with SciPy, Scikit learn for basic machine learning tasks.

Unit 9: Web Scraping and HTML Parsing

Analyzing a web page, Requests, HTML Parsing, BeautifulSoup, Selenium for browser automation and scraping dynamic content, Interacting with forms, third-party APIs, Sequential and Threaded Crawlers, Concurrent Downloading, Caching, Solving CAPTCHA, Scrapy for Web Scraping, RSS feeds.

Unit 10: FastAPI

Request-response model, HTTP methods: GET, POST, PUT, DELETE, PATCH, HTTP status codes, RESTful API Concepts, Pydantic class, create your own api. Postman GUI.

Unit 11: Dockerization

Introduction to Docker, Key Concepts: Image, Container, Dockerfile, Registry, Installing Docker and verifying setup, Writing a simple Dockerfile for a Python project, Building and running Docker images, Managing containers (start, stop, logs, exec), Docker volumes for persistence, Docker networking basics, Publishing and pulling images from Docker Hub, Best practices for lightweight images.

Unit 12: GUI Programming: Introduction to GUI Development with Streamlit and Gradio

Gradio: Installation, Creating Basic Interfaces, Layouts and Components, Event Handling and Interactivity (Using submit, change, and click events, Connecting multiple functions with components, Updating outputs dynamically), Hosting Gradio apps with share=True, Practical example, Streamlit: Installation, Creating and running basic

apps, Using `st.write()`, `st.text()`, `st.markdown()`, `st.header()`, `st.subheader()`, Showing data with `st.dataframe()` and `st.table()`, `TextInput`, Widgets (text, selectbox, radio, checkbox, file etc.), Interactivity and Events (button and `session_state`), Practical example.

Unit 13: MongoDB in Python:

Python using `pymongo`, creating databases and collections, performing CRUD operations including inserting, querying, updating, and deleting documents, filtering data with comparison and logical operators, projecting specific fields, sorting query results, creating basic indexes, using simple aggregation pipelines for grouping and summarizing data, and practical exercises to reinforce these core concepts.

02 SQL (Structured Query Language)

[10 weeks]

Course Learning Objectives

Upon completion of this curriculum, interns will be able to:

- Understand SQL, relational database fundamentals, and modern database architectures.
- Execute SQL queries for data creation, manipulation, and retrieval.
- Use filtering, aggregation, joins, and subqueries for analytical data operations.
- Implement views, stored procedures, transactions, and triggers for real-world applications.
- Apply query optimization and indexing for better performance.
- Integrate SQL concepts with data analytics and visualization workflows.

UNIT 1: Introduction to Databases and SQL Fundamentals

Introduction to Databases and RDBMS (MySQL, PostgreSQL, SQL Server), Types of Databases: Relational vs Non-Relational (SQL vs NoSQL overview), Database Architecture: Tables, Rows, Columns, Schema, Constraints, SQL Syntax and Query Structure, Data Types: Numeric, String, Boolean, Date/Time, DDL (Data Definition Language): CREATE, DROP, ALTER, TRUNCATE, DML (Data Manipulation Language): SELECT, INSERT, UPDATE, DELETE, DCL (Data Control Language): GRANT, REVOKE, TCL (Transaction Control Language): COMMIT, ROLLBACK, SAVEPOINT, Basic SQL Clauses: SELECT, WHERE, ORDER BY, DISTINCT, LIMIT, KEYS: PRIMARY KEY, FOREIGN KEY, UNIQUE KEY, COMPOSITE KEY, Constraints: NOT NULL, DEFAULT, CHECK, ACID Properties: Atomicity, Consistency, Isolation, Durability — with practical examples, Normalization: 1NF, 2NF, 3NF (Introductory Concept).

UNIT 2: SQL Data Retrieval, Filtering, and Functions

Retrieving Data from Tables, Filtering Data: WHERE, LIKE, IN, BETWEEN, IS NULL/IS NOT NULL, Sorting and Limiting Results: ORDER BY, LIMIT, OFFSET, Aggregate Functions: COUNT, SUM, AVG, MIN, MAX, Grouping Data: GROUP BY, HAVING, Joins (Data Integration): INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL OUTER JOIN, CROSS JOIN, SELF JOIN, ANTI JOIN, Subqueries and Nested Queries (in SELECT, FROM, WHERE, INSERT, UPDATE, DELETE), SQL Functions: Scalar Functions (UPPER(), LOWER(), ROUND(), LENGTH()), String Functions (CONCAT(), SUBSTRING(), REPLACE(), TRIM()), Date & Time Functions (NOW(), CURDATE(), DATEDIFF(), DATE_FORMAT()), Mathematical Functions (ABS(), CEIL(), FLOOR(), MOD()).

UNIT 3: Advanced SQL Operations and Optimization

Set Operations: UNION, UNION ALL, INTERSECT, EXCEPT, Conditional Logic: SQL CASE Statement, Advanced Filtering Operators: EXISTS, ANY, ALL, Working with Multiple Tables and Complex Joins, Views: Creating and Using Views for Query Simplification, Indexes: Types (Clustered, Non-Clustered), Benefits, and Use Cases, Query Optimization: EXPLAIN / ANALYZE for Query Execution Plans, Index Optimization, Avoiding Subquery Overheads, Transactions: COMMIT,

ROLLBACK, SAVEPOINT — Use in Real Scenarios, Stored Procedures and Functions: Creation, Execution, and Use Cases, Triggers: Event-Based Automation in Databases, Error Handling and NULL Management.

UNIT 4: Applied SQL for Data Analysis and Integration

Database Design & Relationships: One-to-One, One-to-Many, Many-to-Many, Data Modeling for Analytics, Working with Real-World Datasets (e.g., Perfume Store, E-Commerce, IoT Data), Common Table Expressions (CTE) and Recursive Queries, Pivoting and Unpivoting Data, Integration with Analytics Tools:, Using SQL in Python (via Pandas/SQLAlchemy), SQL with Business Intelligence tools (Power BI, Tableau overview), Data Export & Import: CSV, JSON formats, Best Practices: Naming Conventions, Query Formatting, Documentation.

03 Applied Machine Learning [18 weeks]

Course Learning Objectives

The objectives of this course is to make students to

- Understand the fundamental concepts of supervised and unsupervised machine learning.
- Develop skills in data preprocessing, feature engineering, and model evaluation.
- Implement classification, regression, clustering, and dimensionality reduction techniques.
- Learn to assess model performance using appropriate metrics.
- Apply machine learning concepts to real-world problems using Python.

UNIT 1: Foundations of Machine Learning & Data Preprocessing

Mathematical & Statistical Foundations: Descriptive Statistics (Mean, Median, Mode, Variance, Standard Deviation), Probability Concepts (Distributions, Functions, and Random Variables), Hypothesis Testing & P-Values, Matrices and Vector Representations, Data Preprocessing Techniques; Data Scaling (Min-Max, Standardization, Normalization),

Encoding Categorical Data (One-Hot Encoding, Dummy Variable Encoding), Introduction to Machine Learning: What is Machine Learning?, Types of ML (Supervised, Unsupervised, and Reinforcement Learning), Real-world Applications of ML.

UNIT 2: Supervised Learning – Applied

Regression Techniques: Linear Regression, Model Assumptions, Model Evaluation for Regression (MSE, RMSE, MAE, R^2 Score), Classification Techniques, Logistic Regression (Binary & Multiclass), Naïve Bayes Classifier, Decision Trees, K-Nearest Neighbors (KNN), Model Evaluation for Classification: Confusion Matrix, Precision, Recall, F1-Score, ROC Curve & AUC, Model Optimization & Generalization: Bias-Variance Tradeoff, Overfitting & Underfitting, Cross-Validation Technique.

UNIT 3: Advanced Supervised Learning – Applied

Regularization Methods: L1 (Lasso Regression), L2 (Ridge Regression), Elastic Net, Support Vector Machines (SVM): Concept of margins and support vectors, Linear and non-linear kernels, Ensemble Learning (Variance Reduction – Bagging): Bagging concept, Random Forest, Boosting Methods (Bias Reduction – Sequential Models): Gradient Boosting Machines (GBM), AdaBoost, XGBoost, Linear Discriminant Analysis (LDA).

UNIT 4: Unsupervised Learning & Dimensionality Reduction – Applied

Clustering Algorithms: K-Means Clustering, Hierarchical Clustering, DBSCAN, Expectation-Maximization, Dimensionality Reduction Techniques: Principal Component Analysis (PCA), Singular Value Decomposition (SVD), Factor Analysis, Probabilistic & Hidden Variable Models: Gaussian Mixture Models (GMM), Latent Dirichlet Allocation (LDA, topic modeling), Hidden Markov Models (HMM).

04 Power BI

[8 weeks]

Course Learning Objectives

The objectives of this course is to make students to

- Learn how to transform complex business data into actionable insights using Power Query and DAX.
- Understand how to design professional dashboards that align with corporate branding and visualization standards.
- Develop the ability to automate reporting workflows to increase productivity and reduce manual effort.
- Gain practical experience in solving real-world business problems such as sales analysis, performance tracking, and forecasting.
- Apply best practices in dashboard design to enhance clarity, usability, and data storytelling effectiveness.

UNIT 1: POWER BI BASICS – SETUP, DATA TRANSFORMATION & VISUALIZATIONS

Power BI Basics: Introduction, Installing Power BI Desktop, Configuring Initial Project Settings, Using Power Query Editor for Data Transformation, Fixing Data Source Errors in Power BI, Power BI Data Transformation, Changing Column Data Types for Accurate Analysis, Changing Data Types Using Locales, Data Cleaning, Visualizations & Data Modeling: Extracting Values, Optimizing Data Preparation, Cleaning and Transforming Text Data, Numerical Transformations, Building Efficient Relationships for Dashboards, Visualization Techniques, Formatting, Data Labels in Power BI Charts, Using Pie Charts Effectively, Data Preparation, Modeling and Visualization: Appending Queries in Power BI, Grouping and Managing Queries, Hiding Tables, Using Date Hierarchy in Line Charts for Time-Based Visualizations, Drill Down Mode, Creating and Customizing Line Charts, Combining Multiple Files from a Folder, Create a Fact-Dimension Model, Understanding Cardinality, Managing Active and Inactive Relationships in Data Models, Data Transformations: Power BI Table Visualization, Formatting Tables, Conditional Formatting Rules for Data Visualization, Merge Queries for Comprehensive Dashboards, Pivot & Unpivot Techniques for Analytics, Power BI Relationships.

UNIT 2: INTERMEDIATE VISUALS, FILTERS & CUSTOM VIEWS

Interactive Visualizations: Power BI Slicers, Reports Designing, Filters, Advanced Filtering Techniques, Slicer Sync, Treemap Visuals, Edit Interactions, Drill Through, Tooltips, Advanced Transformations & Visualizations: Using M Language, Enabling and Disabling Load, Query Management, Optimizing Power BI Charts, Creating a Conditional Column, Creating Interactive Map Visualizations, Filled Maps, Line Charts, Anomaly Detection, Dashboard Enhancements: Creating Interactive Drillthrough Buttons, Clear and Apply Slicer Buttons, User-Friendly Buttons, Creating and Using, Card Visual, Multi-Row Cards, Custom Tooltips, Theme Gallery.

UNIT 3: DAX, TIME INTELLIGENCE & CALCULATIONS

DAX essentials includes Creating Calculated Columns in DAX, Measures vs Calculated Columns, Using DAX Average Function, Creating a Measure Table, DAX Basics, SUMX and AVERAGEX, Row-Based Functions in DAX, DAX Essentials, DAX- CALCULATE & Filtering and Time Intelligence: DAX Filter Context, FILTER Function in DAX, DAX Logical Operators, Overcoming Filter Context with ALL Function, Optimizing Percentage Calculations with ALL Function, ALLEXCEPT Function in DAX, ALLSELECTED Function in DAX, DATEADD Function in DAX for Time Intelligence, Running Totals with DAX, DAX FORMAT Function, Time Intelligence.

UNIT 4: POWER BI CLOUD, SECURITY & ADVANCED INTEGRATION

Publishing reports & Power Bi service: Accessing and Consuming Reports Anywhere in the World, Understanding Power BI and Fabric Subscriptions, Understanding Power BI Service and Microsoft Fabric, Sign Up for Power BI Service and Microsoft Fabric, Power BI Service Setup, Setup Free Trial for Fabric Premium Capacity, Creating a Power BI Workspace with Premium Capacity, Publishing Reports and Semantic Models to Your Workspace, Working in Power BI Service and Editing Reports, Understanding Semantic Models, Power BI Report Management, Mastering Power BI Workspace Access, Scheduling Data Gateway & Mobile App: Creating Apps, Schedule Refresh and Dashboards, Optimizing Report, Creating and Distributing Power BI Apps, Updating Existing Apps, Updating Changes in Underlying Data,

Set Up Power BI Gateway for Automatic Data Refresh in Power BI Service, Set Up Scheduled Refresh, Set Up and Use Power BI Mobile App, Creating Mobile-Friendly Power BI Reports, Row-Level & Page-Level Security (RLS): Creating Custom Roles for Data Access Control, Publishing and Testing Security Measures in the Cloud Workspace, Understanding Data Model Relationships for Effective RL, Setting Up Dynamic RLS with External Tables and DAX, Fixing Many-to-Many Relationships in RLS, Implementing Page-Level Security in Power BI , Python, Machine Learning & SQL: Integration of Python with Power BI for Advanced Data Analysis, Installing Python and Connect It with Power BI, Using Python in Power BI, Using Python Visuals in Power BI, Creating Violin Plots Using Python & Seaborn for Data Analysis, Troubleshooting Python Integration in, Connecting Power BI to REST API, Setting Up MySQL Workbench for Power BI Integration.

05 PySpark

[7 weeks]

Course Learning Objectives

The objectives of this course is to make students to:

- Understand PySpark architecture and components, including Driver, Executors, Cluster Managers, and SparkSession setup.
- Develop proficiency in RDD operations—creating, transforming, aggregating, and managing distributed datasets.
- Gain hands-on experience with DataFrames and SQL in PySpark for structured data manipulation, querying, and transformation.
- Learn data cleaning and preprocessing techniques using PySpark functions for handling nulls, duplicates, joins, and nested data.
- Explore machine learning with MLlib, including feature engineering, model building, and pipeline creation for large-scale data analysis.

Unit 1: Introduction to PySpark

What is PySpark & Why Use It, PySpark Architecture includes Driver, Executors, Cluster Managers, SparkSession & SparkContext, Setting up PySpark, Hands-on includes Create first PySpark Session & run basic operation.

Unit 2: Core RDD Operations (Part 1)

RDD Concepts includes Lazy Eval, Fault Tolerance, Immutability, Creating RDDs (parallelize, textFile), Basic Transformations: map(), flatMap(), filter(), Actions includes collect(), count(), reduce(), Hands-on includes Word count, text cleanup, line parsing.

Unit 3: Core RDD Operations (Part 2)

Key-Value RDDs includes groupByKey(), reduceByKey(), sortByKey(), More ops includes union(), distinct(), glom(), RDD Partitions & repartition(), Hands-on includes Aggregate sales by product, repartition a dataset.

Unit 4: DataFrames - The Structured API

What are DataFrames & How they differ from RDD, Schemas (automatic vs manual), Creating DataFrames (lists, CSV), Basic Ops includes select(), filter(), withColumn(), Hands-on includes Load CSV → Clean → Select → Filter.

Unit 5: SQL with PySpark

Creating Temporary Views, Running SQL Queries (SELECT, WHERE, ORDER BY), SQL vs DataFrame API, Hands-on includes Querying CSV via SQL (top N, group by, sorting).

Unit 6: Data Cleaning in PySpark

Handling nulls includes dropna(), fillna(), Dropping duplicates, Renaming columns, Type conversions includes cast(), Hands-on includes Clean messy customer dataset.

Unit 7: Advanced DataFrame Transformations

withColumn() to modify/add columns, Joins includes Inner, Left, Right, Outer, Working with nested data & arrays, Using explode() for flattening, Broadcast join (intro), Hands-on includes Customer → Order join, explode hobbies array.

Unit 8: Introduction to Machine Learning with PySpark (MLlib)

What is MLlib (Machine Learning Library in PySpark), Basic ML

Concepts includes Training, Testing, Models, Supervised Learning (Classification, Regression), Feature Engineering includes StringIndexer, VectorAssembler, Normalizer, Intro to Pipelines in PySpark, Hands-on includes Preprocessing data for ML (load data → feature creation).

06 Dataiku

[7 weeks]

Course Learning Objectives

The objectives of this course is to make students to:

- Gain proficiency in Dataiku platform fundamentals, including visual and code-based data preparation, exploration, and pipeline design.
- Develop skills in machine learning and advanced analytics, covering predictive modeling, explainable AI, NLP, image classification, and time series forecasting.
- Learn to implement Generative AI workflows with LLMs and Retrieval-Augmented Generation (RAG) within Dataiku.
- Build expertise in developer and coding practices, including plugin development, custom ML models, API integration, and Git-based project management.
- Apply MLOps and governance practices to deploy, monitor, and govern AI/ML solutions efficiently in production environments.

UNIT 1: Core Designer Certification

Interface & Data Exploration includes Course Overview, Discover the Interface, Create the Project, Create the Dataset, Explore Your Data, Visual Recipes includes Introduction to visual recipes, Preparing Data, Joining Data, Aggregating data, Reordering data, Transforming Data, Reducing Data, Unioning and Separating Data, Moving Data, Collaboration, Dataiku & SQL includes Connect to SQL Databases, Move Data to Your Database, In-Database Data Visualization & Preparation, SQL Notebooks & Recipes, Geospatial Analytics.

UNIT 2: Advanced Designer

Variables, Data Pipelines, Data Quality & Automation, Dataiku Applications, Git for Projects Partitioning includes Partitioning Concepts, Partitioning Redispatch, Partitioning in a Scenario, Project Deployment includes Preparing for Deployment, Batch Deployment.

UNIT 3: ML Practitioner

Machine Learning Basics includes Create the Model, Evaluate the Model, Tune the Model, Explainable AI, Scoring Basics includes Deploy the Model, Scoring Data, Interactive Statistics includes The Interactive Statistics Interface, Univariate and Bivariate Analysis, Fit Curves and Distributions, Correlation Matrix, Principal Component Analysis, Statistical Testing, Generate Statistics, Intro to Machine Learning includes Introductory Concepts, Predictive Modeling, Prediction includes Regression & Classification, Clustering, Machine Learning 102 includes Model Development Strategies, Model Diagnostics, Actionability, Responsible AI includes Introduction to Responsible AI, Identifying Data Biases, Model Fairness & Interpretability, Model Explainability & Reporting, Partitioned Models, NLP - The Visual Way includes Introduction to NLP, Preparing Text Data, Handling Text Features for ML, Image Classification and Object Detection without Code includes Image Classification, Object Detection, Time Series Analysis & Forecasting includes Time Series Basics, Time Series Analysis, Time Series Forecasting, Time Series Preparation includes Time Series Preparation, Resampling, Interval Extraction, Windowing, Extrema Extraction, Decomposition.

UNIT 4: Generative AI Practitioner

Text Processing With LLMs Using Visual Recipes includes Introduction to Large Language Models (LLMs), LLM Recipes, Retrieval Augmented Generation (RAG) with LLMs includes Embed recipe and Retrieval Augmented Generation (RAG).

UNIT 5: Developer

Code within Dataiku includes Introduction, Core Coding Interface, Code Environments, Variables for coders, Static insights, Managed Folders, The Public API, Custom Automation includes Custom metrics, checks, data quality rules, & scenarios, Plugin Development includes

What are Development Plugins?, Developing Plugins, Git Integration for Plugins, Hands-On includes Plugin Development, Dataiku for R Users, Webapps, Custom ML Models includes Custom Preprocessing in Visual ML, Custom Modeling in Visual ML, Shared Code includes Introduction, Libraries, Importing from Git, Code Samples.

UNIT 6: MLOps Practitioner

Production Concepts includes Intro to MLOps, Project Deployment includes Preparing for Deployment, Batch Deployment, API Deployment, Production Monitoring includes Production Monitoring Foundations, Model Monitoring with a Model Evaluation Store, Model Monitoring Feedback Loops, Production Deployment Automation, Model Monitoring in Different Contexts, Dataiku Govern includes Centralization, Initializing Governance, Tracking Progress, Deployment, Advanced Dataiku Govern includes Introduction, Blueprint Designer, Custom Pages Designer.

07 Applied Large Language Models

[19 weeks]

Course Learning Objectives

The objectives of this course is to make students to

- Understand the fundamental concepts of CNN and Transformer
- Understand and implement LLM and RAG based application
- Discover and solve challenges in hosting and evaluation of LLM applications.
- Understand research trends in LLM

UNIT 1: Review Convolution Neural Network

Basics of NLP and applications, Introduction to embeddings and their role in modern LLMs, Text preprocessing steps (tokenization, stop-word removal, stemming, lemmatization), Text representations & vectorization (Bag of Words, TF-IDF, Word2Vec, GloVe), Neural Network Fundamentals Refresher – perceptrons, activation functions, loss functions, optimization, Sequence Modeling Basics – RNNs, LSTMs, GRUs, and why they were precursors to Transformers.

UNIT 2: Transformers

Transformers in action: Simple capabilities of transformers using pretrained models and pipeline, Exploring Hugging Face for different transformer models for different uses, Types of Transformers (BERT, GPT comparisons), Mechanics of Seq2seq Models With Attention, Transformer architecture: self attention, multihead attention, Query-Value_Key triplet, Encoding Strategies, Fine Tuning Techniques, Zero-Shot Learning, One-Shot Learning and Few-Shot Learning.

UNIT 3: Large Language Models

Introduction to LLMs (Overview of LLMs, Types of LLM Models, Open-Source Models): Model Architectures & Variants (Mixture of Experts (MoE) Models, Vision Models, Multimodal LLMs): Infrastructure & Resources (Memory Management, GPUs/TPUs): Training & Optimization (RLHF for LLMs, Fine-tuning – Dataset Preparation, Infrastructure and Monitoring, LoRA, PEFT, and Parameter-Efficient Tuning): Deployment & Hosting (Hosting Open-Source LLMs, Hosting in Hugging Face and other Cloud Networks).challenges involved in Bias in LLM outputs, Toxicity, misinformation.

UNIT 4: Prompt Engineering and RAG

Few-shot, zero-shot, and chain-of-thought prompting, System vs. user prompts, Prompt design tools and prompt libraries, Prompt optimization and evaluation frameworks, Retrieval Augmented Generation (RAG), Semantic search , Fine Tuning Ranking model, Search based and Graph based retrieval.

UNIT 5: Agentic Network

Understanding Tools and Agents, Study of agentic frameworks like Autogen and Langgraph, Multiagent network creation and use cases, challenges and limitations.

08 Capstone Project

[9 weeks]

At the culmination of the program, students will design and implement a real-world Data Analytics project, applying knowledge and skills gained across all modules. They will begin by using SQL and Dataiku to collect and manage data from diverse sources, while Python and PySpark will be used for large-scale data cleaning, transformation, and exploration. Through Power BI and Python visualization libraries, students will create compelling visual insights to support data storytelling. They will then build and evaluate predictive models using Applied Machine Learning techniques and integrate Applied LLM tools to generate insights from unstructured text or automate analytical reporting. Dataiku will further support workflow automation and collaboration. The project will culminate in a final dashboard and presentation that showcases both technical depth and business understanding, offering a complete end-to-end analytics solution.

Program Outcomes

Upon completion of this program, students will achieve:

1. Mastery of Python for data analysis, visualization, and database management.
2. Ability to clean, process, and analyze real-world datasets for actionable insights.
3. Proficiency in business intelligence tools, SQL, and data-driven decision-making.
4. Understanding of machine learning techniques relevant to data analytics.
5. Hands-on experience with big data tools for scalable data processing.
6. Design interactive dashboards and reports using Power BI for data storytelling and business insights.

Certification

Upon successful completion of Sabudh's Internship Program, you will be awarded a **"Certificate of Completion"**, in Data Analytics. This certification validates your proficiency in data analysis, visualization, machine learning, and real-world problem solving—key skills sought by employers.

Your final portfolio will not only demonstrate technical expertise but also showcase your ability to derive insights, think critically, and apply data-driven strategies to solve complex business challenges.



THANK YOU



SABUDH

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